⇒ Username : surya-506

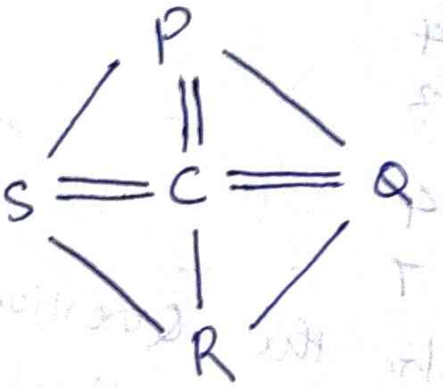
⇒ Enigma number : 3101

$(19 \times 1) + (21 \times 2) + (18 \times 4) + (25 \times 8)$

$+ (1 \times 16) + (0 \times 32) + (19 \times 64) + (0 \times 128)$

$+ (6 \times 256) = 3101$

⇒ Bridges of my city



Last 4 digits : 3101

⇒ My given : The walk is possible

⇒ My attempts :

(1) C → Q, Q → P, P → C, C → P, P → S, S → C, C → S, S → R, R → C, C → Q, Q → R (possible)

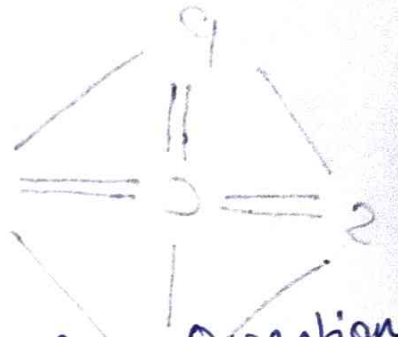
(2) C → S, S → C, C → P, P → Q, Q → C, C → Q, Q → R, R → C, C → P, P → S, S → R (possible)

(iii) (from AI)

$R \to S$, $S \to P$, $P \to C$, $C \to Q$, $Q \to P$,
$P \to C$, $C \to S$, $S \to C$, $C \to Q$, $Q \to R$,
$R \to C$ (possible)

⇒ Bridges touching each Island

Island	No. of bridges
P	4
Q	4
R	3
S	4
C	7



My Answer to the Questions :-

1, 2, 3

What surprised me, what I think and the pattern

→ I noticed that if I start from "C", I'd be able to solve it in 7 different ways; it's kind of like tic-tac-toe where the middle position's the most important one.

But when I asked chatgpt for the answer (to cross-verify) it told me that starting from a vertex

that has an odd degree is the way to go and the underlying concept here is a technique in Hierholzer's algorithm.

I found it funny that the pattern I figured out, is not the norm and graph theory tells you to do the exact opposite.

AI disclosure

I did use AI to cross-verify my answer, know other possible routes and underlying concept on the routes it gave me.